

BLDC Motor Control Algorithm for Low-Cost Industrial Applications

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Abstract - Electrical motors are an integral part of industrial plants with no less than 5 billion motors built word wide every year. The demand for low-cost brushless dc (BLDC) motors have increased in industrial applications. This paper presents a BLDC motor control algorithm for low-cost motor drive applications using general purpose microcontrollers which have only one on-chip timer. The papere describes how to realize pulse width modulation (PWM) signals with general input/output (I/O) ports to control a three-phase permanent magnet brushless dc motor using the timer interrupt on MSP430F1232.

Keywords: BLDC motor, PWM, Industrial application, MSP430F1232.

I. INTRODUCTION

Low end applications of motor control can be found in diverse line of consumer, medical, and industrial products. High efficiency variable speed and variable torque motor control is only possible using electronic components and microcontrollers. Low cost is still dominant factor in designing very low-end products using motor control. Examples are hand-held power tools and home appliances in the market.

Battery powered vacuum cleaner, drill, and electric saw units were investigated. In these products, brush dc or universal motors were used. The torque/speed variation is handled roughly by switching 2-position gearbox unit connected to motor shaft. That is, low cost design has been achieved by transferring the cost of electronic drives into the motor type and mechanical parts (gearbox) sacrificing efficiency and actual performance of the product for the end user applications. Battery powered tools could only operate at their nominal speed/torque for a short period of time before recharging the battery became necessary. As in the case of power drill and saw, small increase of the torque reduced the rotational speed of the motor-controlled system. It is also interesting to note that the price of the battery charger is quite comparable with the price of the final product.

Digital electronic control of motors would offer much higher efficiency and better power usage (longer operating period for the battery) at competitive cost. Despite all the obvious facts, replacing the cost of

inefficient noisy motor types and the mechanical components with that of the more efficient brushless motors and the digital electronics/microcontrollers in home appliances and hand-held power tools (with cord or cordless) would still require concentrated technical support and marketing effort. Reference designs, comparative cost analysis, and application notes for specific motor control end equipment will provide design engineers with the benefits of the microcontroller-based systems shortening learning curve and their design efforts.

A motor control system can be partitioned into 4 units: supply line interface, digital processor, motor and sensors, power converter or drive unit. Electric power is supplied to the motor control system via supply line interface in the form of ac or dc signal. Motors use windings on the stator parts and either permanent magnet (PM) or windings on the rotor parts. Stator windings is either separated or connected to the rotor windings in series or parallel configuration. Mechanical components (brushes) are needed to connect electrical signal to the rotating part. In a motor, rotation of rotor and torque generation occur when the rotor field and stator field are not parallel, and maximum torque is obtained when these fields are orthogonal to each other. Motors with stator windings in series with that of the rotor (using brushes) can operate with either ac or dc signals and are known as universal motors. Despite their very poor efficiency and poor operations (speed versus torque/load, noise, electric sparks, brush wear-out, size), they are still widely used in household appliances, power tools, and yard tools.

The brushless dc motor (BLDC) motor has been used in various industrial applications and has increased demand in diverse fields because of its high efficiency, simple control compared with ac motors, low EMI, and high reliability due to absent of brushes. Most three-phase motors, including BLDC motor need at least six PWM channels for inverter power devices such as IGBTs and MOSFETs. In order to meet these requirements, generally a special-purpose processor or a programmable logic device (PLD) or drive device to generate control signal is necessary. Using special purpose processor or device for BLDC motor drive presents several advantages such as small drive size and less development time. However, these processors are more expensive than the general purpose processors and eventually will increase the cost of BLDC motor drive system.

Since the main flux of BLDC motor is produced by the permanent magnets, this motor has high power density and is capable of operating at high efficiencies while having similar torque control performance as a dc motor [1-2]. Trapezoidal type BLDC motors are generally used for low-cost industrial applications. Therefore, low-cost application of BLDC motors are being considered for many products, especially its sensorless operation [3].

Sensorless techniques for BLDC motor control with trapezoidal back-EMFs can be classified into two categories: algebraic equation-based techniques [4-6] and back-EMF voltage sensing techniques [7-10]. The main idea of using algebraic equations is to calculate the flux linkage from the motor parameters and voltage/current measurements. Based on the estimated flux linkage, the rotor position can be detected. This method can be operated over a wide speed range, but the motor parameters need to be known precisely and at least one current sensor is required. Therefore, the system performance depends on the accuracy of the motor parameters.

The sensorless techniques that utilize the back-EMF voltage of the open-phase include the following methods,

- 1) Terminal voltage sensing,
- 2) Third harmonic back-EMF voltage sensing,
- 3) Freewheeling diode conduction current sensing.

When a phase is open, the back-EMF voltage which includes the information of the rotor position can be measured at the motor terminals. Since the terminal voltage sensing is simple from the hardware point of view, this approach is widely used in industry for sensorless control of BLDC motors.

The purpose of these sensorless methods is elimination of the position sensor, which are generally three hall sensors mounted across from the rotor. Sensorless techniques need extra computation time and external circuitry to estimate the back-EMF than the sensor-based systems. Therefore, sensorless techniques demand high performance processors, large program codes, and large memory. Finally instead of position sensor, other components are required, so it is very difficult to decrease the total motor drive cost.

In order to achieve low-cost BLDC motor drive, a general-purpose processor with hall sensors is used instead of eliminating the position sensor and then uses a special purpose processor. In this paper a general-purpose processor, such as MSP430F1232 which is an ultra low-power 16-bit RISC mixed-signal processor from Texas Instruments (TI) is used for battery powered applications. The developed control algorithm is presented and the general purpose digital I/O port is used to generate the PWM signals. To verify the proposed algorithm the operation of a three-phase BLDC motor with three hall sensors under open-loop speed operation is described in this paper.

II. BLDC MOTOR

Permanent magnet (PM) motors are synchronous motors that have permanent magnets mounted on the rotor, and the armature windings located on the stator. PM motors are categorized into two types. The first type is referred to as PM synchronous motor (PMSM) which has sinusoidal back-EMF shown in Fig. (1-a). The other type has a trapezoidal back-EMF and is referred to as the brushless dc (BLDC) motor shown in Fig. (1-b) [1].

In BLDC motor drives, the polarity reversal is performed by power transistors switching in synchronization with the rotor position. Therefore, the BLDC motor has to use either internal or external position sensor to detect the actual rotor position. Also the rotor position can be estimated without the need for position sensor. However this paper uses three hall sensors to determine the actual rotor position.

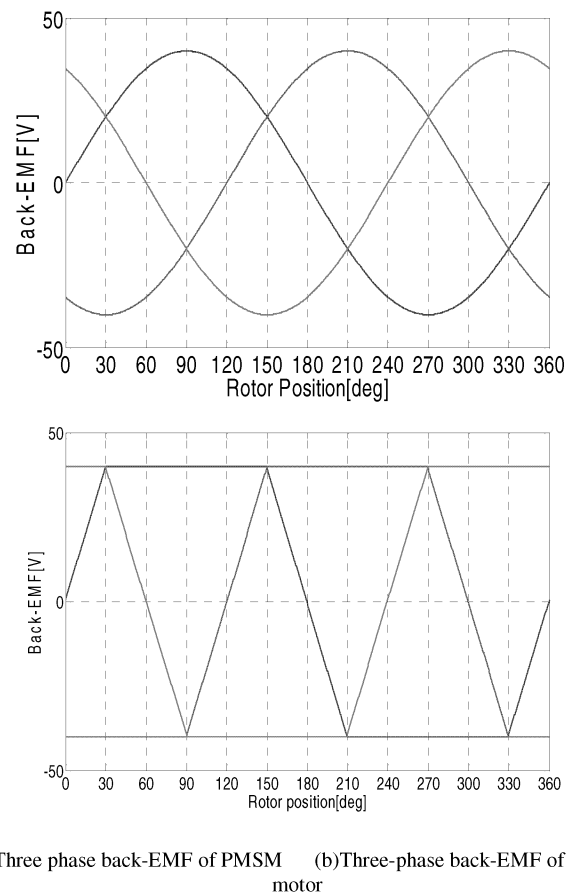


Fig. 1 The back-EMF of PM motors.

In general, BLDC motor may use either 60 deg or 120 deg commutation intervals. In this paper, the 120 degree conduction interval is used. Fig. 2 shows a schematic of BLDC motor and the ideal current waveforms versus position. Also, the position sensors outputs are illustrated. Fig.3 shows the ideal phase current in each of the motor windings and the three-phase inverter, respectively. According to Fig. 3 from the first interval, phase A will conduct positive dc link current while phase B will conduct negative dc link current. Phase C will be left open. As a result, only phase A and B are conducting and

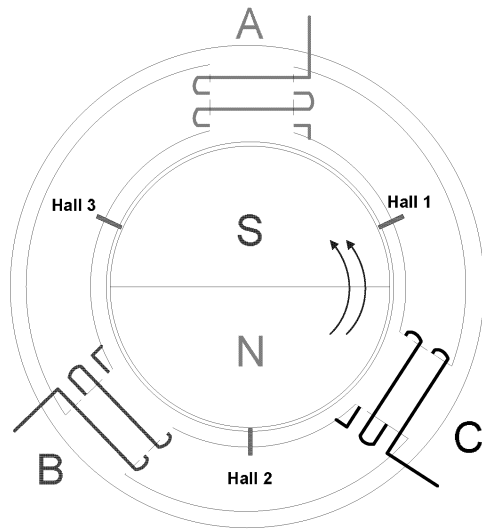


Fig. 2 BLDC motor structure and signals

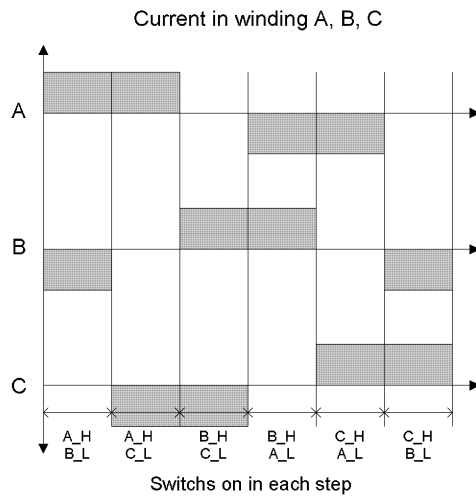


Fig. 3 Inverter configuration of BLDC motor.

phase C is left silent, which means that only two switches (A_H and B_L) are active, while the rest (A_L, B_H, C_H, and C_L) are inactive.

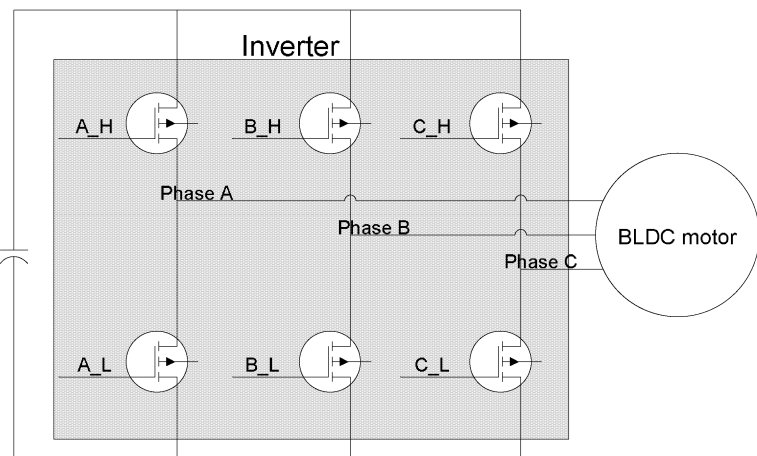
During every PWM cycle to achieve this switching scheme, the desired duty cycle is imposed by the upper switch (A_H) of the two involved switches (A_H and B_L) and the lower switch (B_L) is kept on (100% duty cycle). Therefore, there is no need for dead time to be considered for BLDC motors. Because whenever the upper switch is turned on, the lower switch of the same lag always is already off and vice versa.

III. PROPOSED ALGORITHM

Fig. 4 shows the interrupt block diagram for generating the PWM signal using the I/O port which is called asymmetric PMW strategy. To generate these PWM signals, P3.0-P3.5 ports are used because the MSP430F123 micro-controller does not have enough PWM channels to be directly used with three-phase motor drives and only has two PWM channels. Timer_A has 4 mode, up, down, up/down, and continuous mode, and in this paper, this is configured in up-mode with MCLK as the timer clock source, also Timer_A overflow interrupt and TACCR1 capture/compare interrupt are used to realize exact PWM signal.

In order to keep the frequency constant, the CCR0 register has fixed value, which is 200h, and the CCR1 register has proper value to generate the desired duty ratio. The software checks the value of CCR1 register to ensure that it does not exceed the minimum or maximum value and prevents it from rolling over.

The software flow chart for the three-phase BLDC motor with three hall sensors is described in Fig. 5. The initialization procedure includes the initialization of watchdog timer, digital I/O, Timer_A, ADC10, and variables. The main program consists of the initialization, start of conversion of the dc link current, hall sensor check for rotor position detection, switch signals (SW1, SW2) check for increasing the current reference or decreasing the current reference and the check for motor rotation direction (SW3).



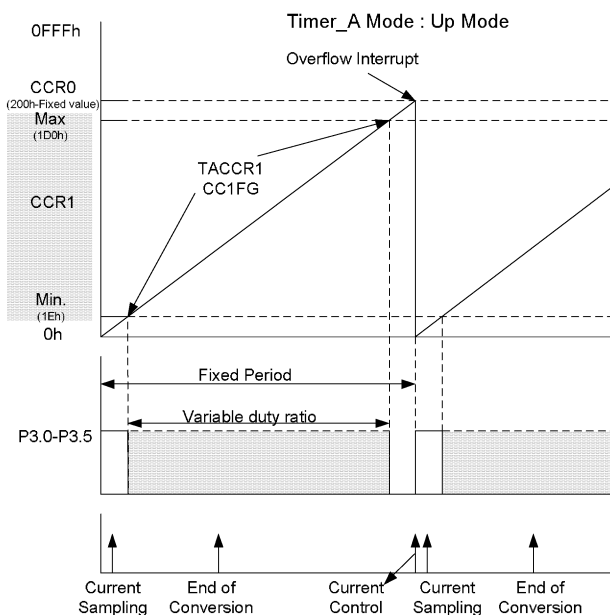


Fig. 4 Timer_A interrupt block diagram for the PWM signal using the digital I/O.

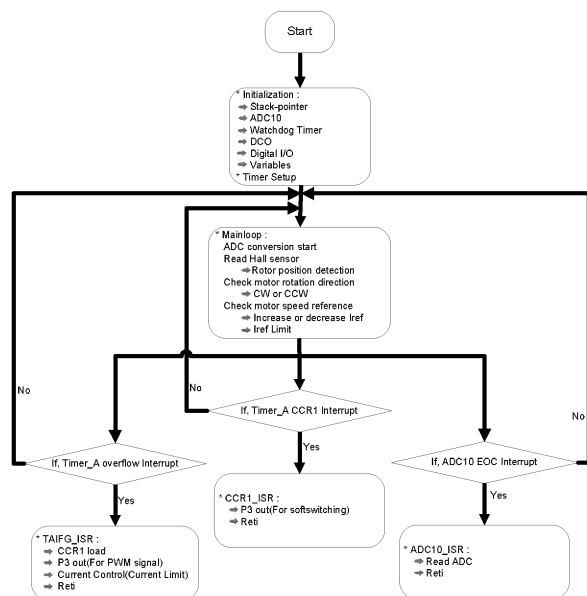


Fig. 5 Software flowchart.

To generate constant PWM signal, P3.0-P3.5 port is used as the PWM output port, and Timer_A underflow interrupt (TAIFG) is served, which is called every 100msec. Also at timer_A underflow interrupt the specific switching pattern for BLDC motor operation is generated based on the position sensors signals. To control the motor current, the dc bus voltage level applied to the motor using proper PWM duty ratio is adjusted accordingly. Timer_A CCR1 interrupt is utilized for regulating this dc bus voltage.

All algorithms are developed in assembly language and it is using 422 bytes of flash memory. The flash memory address is from 0xE000 to 0xFFFF, the interrupt vector from 0xFFFF to 0xFFE0, and the main code memory from 0xFFFF to 0xE000. The developed code is residing in the main code memory. Therefore, there is no need for connecting extra memory and peripheral devices.

IV. SYSTEM CONFIGURATION

Fig. 6 shows the block diagram of the BLDC motor control system used in this paper. MSP430F123 processor receives feedback signal from the dc link current sensor which is used to provide current feedback for the closed loop current control. The rotor position information supplied by the hall sensors of the BLDC motor is estimated with the 3 external I/O port. Actual motor current and the direction of rotation can be changed by the push and toggle switches. Fig. 6 represents the schematic of the BLDC motor drive system. To control motor current, a proportional controller is used to supply proper switching pattern for inverter where three hall sensors are used. As it is shown in this figure to control the BLDC drive system, only MSP430F123 processor is used and external timer devices for PWM generation, memory for program download are never considered. Fig. 7 shows the micro-processor used in this paper.

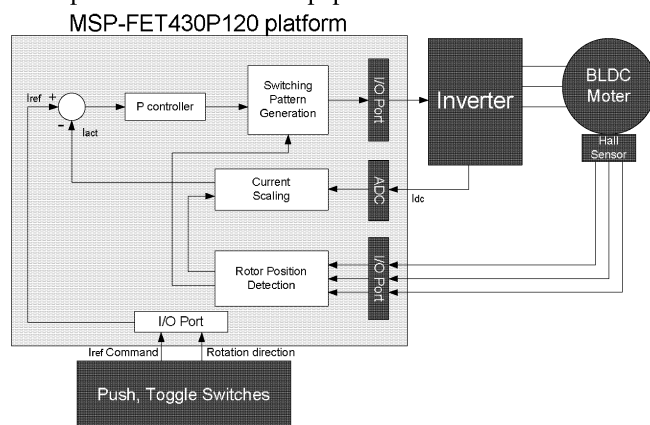


Fig. 6 The schematic of the BLDC motor drive system.

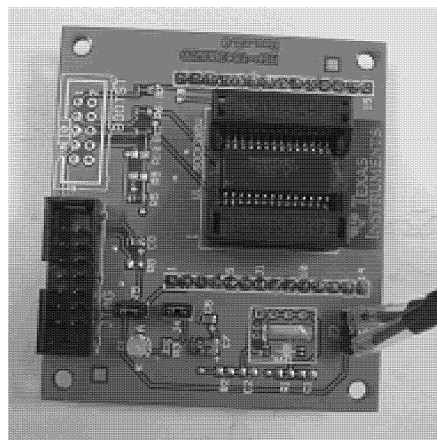


Fig. 7 The MSP430F123 microcontroller board.

Table 1 The comparison of processors for bldc motor drive

Processor	Manufacture	1KU price(us\$)	ADC	Internal Memory	PWM Generation
MSP430F123	TI	2.30 (1 pu)	On-chip ADC (10 bit)	8k flash, 256 RAM	Not need
TMS320F2401A	TI	3.50 (1.52 pu)	On-chip ADC (10bit)	2k RAM	On-chip PWM
ST7MC1K2	ST	3.30225 (1.43 pu)	On-chip ADC(10bit)	16k flash, 4k RAM	On-chip PWM
56F8013	Motorola	3.15 (1.37 pu)	On-chip ADC (12bit)	384-1.5kRAM 8k-60k Flash	On-chip PWM

Table 1 shows different micro-controllers that are commonly used in BLDC motor drive applications. As previously mentioned, MSP430F123 is a general-purpose processor, which means that it does not have enough PWM generators for motor control. Using the developed algorithm for generating the PWM signals instead of the on-chip PWM generator, it is possible to reduce the cost by about 37%. All of these micro-controllers have on-chip ADC and enough internal memory for BLDC motor control algorithm. Therefore, there is no need for extra components to be used for motor currents signal sensing.

V. EXPERIMENTAL RESULTS

Fig. 8 shows the current waveforms, where Channel 1 illustrates the measured dc link current, Channel 2 represents the dc link current waveform using the current probe and Channel 4 indicates the phase current waveform using the current probe.

Fig. 9 shows the PWM signal, where Channel 1 is the A_upper signal, Channel 2 is the B_lower signal and Channel 3 is the C_lower signal using the I/O port and A_phase current waveform using a current probe. As it is seen in Fig.9, the upper leg and the lower leg switches are not on simultaneously. Therefore, there is no need for dead-time to be included in this example.

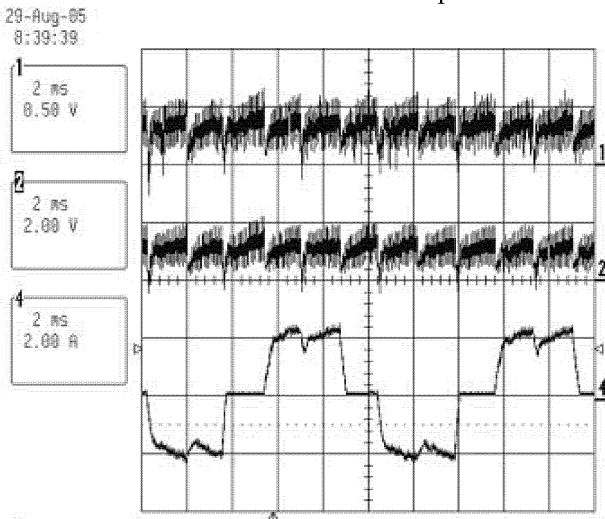


Fig. 8 DC link current and phase current

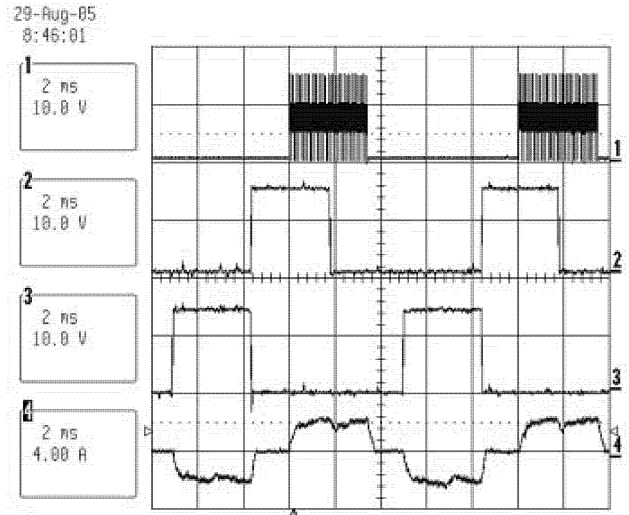


Fig. 9 PWM signals and current waveform

VI. CONCLUSION

In this paper, an efficient algorithm that employs the I/O port for PWM signal generation for BLDC motor control with three hall sensors is proposed. To control bldc motor don't need to be considered dead-time for PWM generation, which allow in proposed algorithms using one timer and general I/O port. The general purpose processor, Texas Instruments MSP430F123 without any peripheral devices is used, to verify the proposed algorithm the control algorithm is implemented on an experimental set up and several tests have been performed. It is shown that the proposed low-cost system has satisfactory performance for industrial applications.

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